Randomized and low-rank approximation

Optimal matvec query complexity of HODLR approximation

Difficulty: challenging **Importance:** interesting to the community

Status: open; checked 2026-09-08
Area: information complexity of hierarchical matrix approximation

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^{\mathsf{T}}v$ for one chosen real vector v; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm's randomness.

Problem statement

Let $n = 2^L k$ with integers $k \geq 1, L \geq 2$, and $0 < \varepsilon < 1/2$. Define $\mathcal{H}_{n,k}$ recursively: a matrix of order at most k is unrestricted; at a larger node, the two equally sized diagonal blocks must satisfy the same definition, and each off-diagonal block must have rank at most k.

Let $q_*(n, k, \varepsilon)$ be the smallest worst-case number of queries with which a randomized adaptive algorithm, given only oracle access to an arbitrary $A \in \mathbb{R}^{n\times n}$, returns $B \in \mathcal{H}_{n,k}$ such that

$$\Pr\left[\|A - B\|_F \leq (1 + \varepsilon) \min_{H\in\mathcal{H}_{n,k}} \|A - H\|_F\right] \geq 0.99$$

for every A. Here only oracle calls are charged: arbitrary measurable computations between queries are allowed. This is an information complexity question.

Question

Determine $q_*(n, k, \varepsilon)$ up to universal multiplicative constants, including its joint dependence on k, L, ε.

Reference

Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson, *Near-optimal hierarchical matrix approximation from matrix-vector products*, SODA 2025, 2656–2692, Definition 1.1, Problem 1.1, Theorems 1.1–1.2, end of §6, and §8.

The known upper bound is $O(\min\{n, kL^4/\varepsilon^3\})$. The source proves $\Omega(kL + k/\varepsilon)$ in its regime $n \geq c_0 k/\varepsilon$ for a sufficiently large absolute c_0; this restriction matters near the full-recovery threshold.

Status evidence

The latest arXiv version is v2, October 24, 2024. Searches for "HODLR query complexity 2026" and "HODLR approximation 2026 lower bound" found no matching improvement. Musco's February 2026 ICERM slides, numbered slide 14, retain the stated upper bound. No later explicit open-status confirmation was located.